

Rhys Talley

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EDUCATION

Columbia University, School of Engineering and Applied Science

Expected May 2027

M.S. in Engineering: Computer Science, Software Systems

Columbia University, School of Engineering and Applied Science

May 2026

B.S. in Engineering, Magna Cum Laude: Computer Science

Cumulative GPA: 4.08 / 4.00

Tau Beta Pi • Upsilon Pi Epsilon

Selected Coursework: Operating Systems, Computer Architecture, Advanced Systems Programming, Design With C++, Computer Networks

WORK EXPERIENCE

Software Developer Intern, IBM Corporation

May 2025 – Aug 2026

- Built applications within IBM's proprietary z/TPF operating system across its Middleware & Security and Database teams, using C++, C, and z/Architecture Assembly (HLASM).
- Implemented compliance tooling to store and monitor remote TLS certificates, hooking into SSL handshake to capture client and server certificates and invoking a client-defined user exit once daily per certificate falling within a client-set expiry window.
- Redesigned capture path when a federal agency client's employee logins resulted in 70,000+ daily connections against a 1,000-entry table, creating a second user exit before insertion occurs, allowing clients to filter certificates using customizable criteria.
- Developed a migration routine to convert hierarchical, record-oriented TPFDf databases into an indexed, compressed structure, improving read times by up to 96%; extended the routine with a fallback path to revert migration for safe rollback.
- Fixed a defect causing shared system sockets to hang until system IPL (restart) – often weeks apart on high-volume clients – by tracking each socket's address in ECB storage and releasing any still held at ECB exit.
- Expanded on driver code for internal HTTP protocol stress testing with size-based header parameters, letting users test up to 1,000,000 bytes each of request and response headers, an increase of 10,000x.

Software Engineering Intern, Columbia Build Labs

Sep 2024 – May 2025

- Designed data models, 9-table SQL schema, and Flask REST API for a property listing platform spanning users, agents, and administrators; data model was carried forward into production.
- Oversaw logic for remote-tour scheduling, agent availability, and inventory management; shipped a QR code waitlist and email-confirmation, collecting 50+ signups ahead of launch.

Teaching Assistant, Computer Science, Columbia University

Jan 2024 – Present

- Taught 1,000+ students across five courses – C++ for C Programmers, Design for Generative AI, Advanced C, Data Structures, and Intro to CS – through office hours and discussion forums; promoted to Head TA for C++ for C Programmers in September 2026.
- Conceptualized grading rubrics and frameworks in Pygrader; drafted exam questions, assignments, and course projects with 40+ co-TAs.

PROJECTS

Static Linker – C, ELF, x86-64

- In a two person team, made a linker capable of merging an arbitrary number of ELF object files into a runnable static executable, generating an ELF header, program and section header tables, symbol table, and string tables from scratch.
- Added global symbol resolution across object files with duplicate- and undefined-symbol detection, then applied PC-relative relocations (R_X86_64_PC32/PLT32) to patch call sites to resolved runtime addresses.
- Tracked file offsets and load addresses separately to lay out page-aligned loadable segments; validated executables against readelf and objdump.

“MyMake” Automation Tool – C++17

- Produced a subset of GNU make capable of parsing a “MyMakefile” into a rule graph using std::regex, resolves variable expansion, and generates implicit compile and link rules from source-file discovery..
- Implemented recursive dependency resolution with cycle detection and timestamp-based staleness checks, dispatching commands through /bin/sh and propagating exit status.
- Wrote a templated serialization library (XdbReader<T> / XdbWriter<T>) and a custom exception hierarchy over std::exception; cached parsed rule graph to disk to skip re-parsing on subsequent builds.

SKILLS

Languages: C++, C, Python, Java, SQL, Assembly (x86-64, MIPS, z/Architecture HLASM)

Systems & Tools: z/TPF, TPFDf, TPF Toolkit, GNU Make, GDB, Valgrind, readelf/objdump, Git

INTERESTS

OS and systems programming, video games, weightlifting, soccer, fantasy novels.